

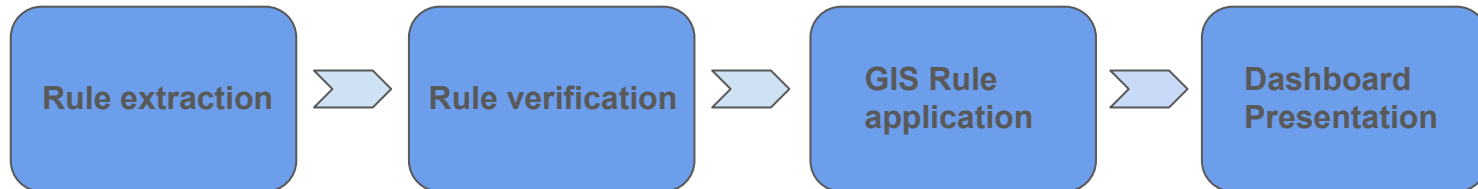
Automated Building Validator - Week 6

Green Metrics Technology (GMT)

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Yusen (Thomas) Rong
Zihao Sheng

Current Project Outline

1. Rule Extraction Pipeline that extract and normalize all the rules from PDF bylaw files (Burnaby, Calgary, Vancouver)
2. Verification Layer to verify all the extracted rules and send the verified ones to GIS layer (Burnaby, Calgary, Vancouver)
3. The GIS layer take in the rules and apply them to city lots (Burnaby and Calgary)
4. The final product will be shown in a dashboard with the GIS map. (Burnaby and Calgary)



Rule Extraction Layer: Zihao

Update since last week:

GMT asked for rule extraction from a new city Calgary, with 1053 pages of bylaws (cut to 6060 blocks) in one pdf file:

1. The rule extraction requires a more precise way to filter unrelated pages and blocks with code

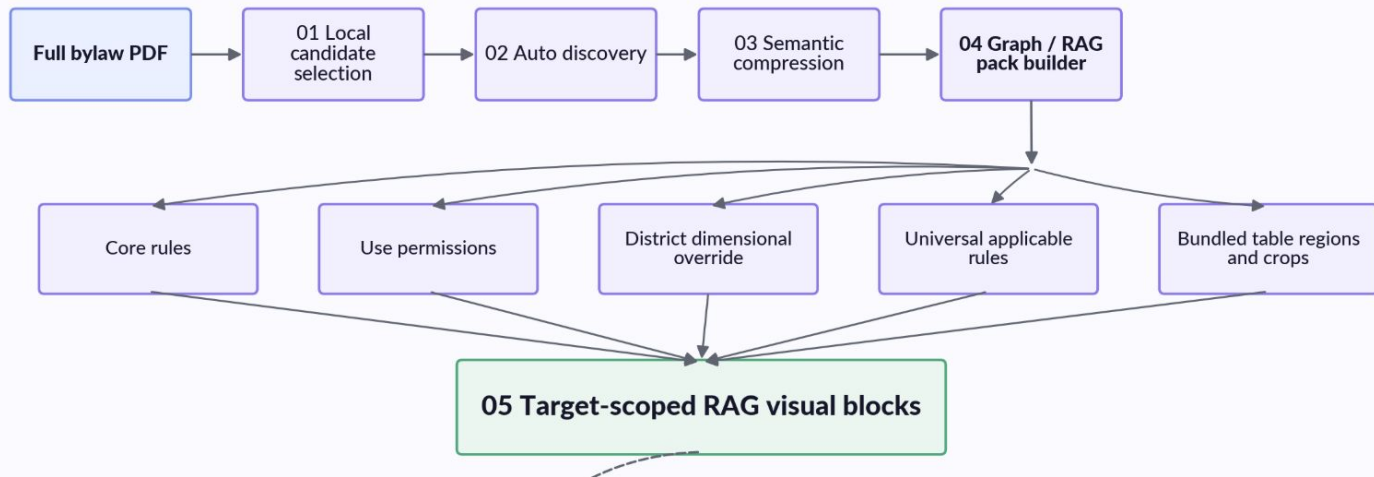
2. Even with filtering, the remained 160 blocks still cost too much token.
Therefore, we need a new API call algorism

Rule Extraction Layer: Zihao

Main upgrades:

1. Advanced pdf decomposition and RAG-like block reconstruction
2. More efficient code-based rule filtering algorithm to reduce unrelated information
3. Batched extraction with caching — reducing 160 API calls to 26 calls. Also now support parallel API calling to save time.

1053 pages of bylaws extracted in less than 5 minutes with 0.3 CAD cost.



[pipeline 10 readme](#)

Rule Verification: Yusen

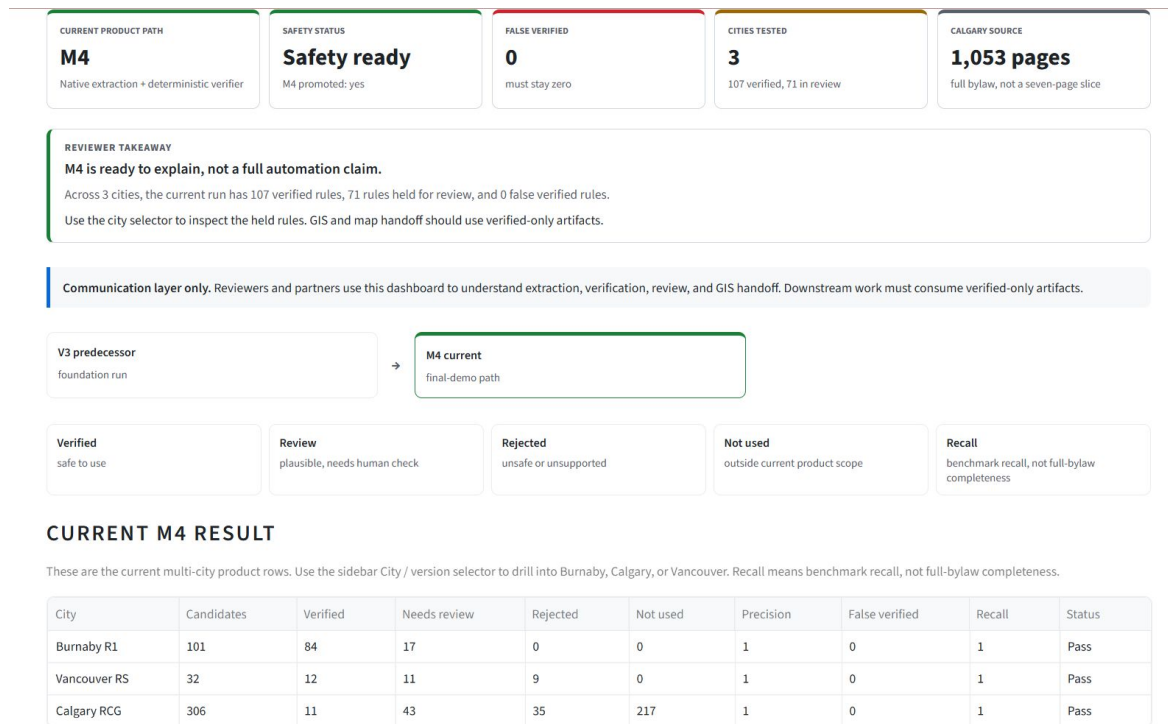
Update since last week:

Verification now runs on Burnaby, Vancouver, and Calgary with zero false verified rules across all three:

1. "Not used" = real rules but outside configured target sections — kept for audit, not exported to GIS
2. Streamlit dashboard tracks quality metrics across all cities
3. Candidate-evidence comparison + evidence repair now available

Rule Verification: Yusen

M4 model:
Reduce review cases
only when benchmark
scores prove it's safe
to do so

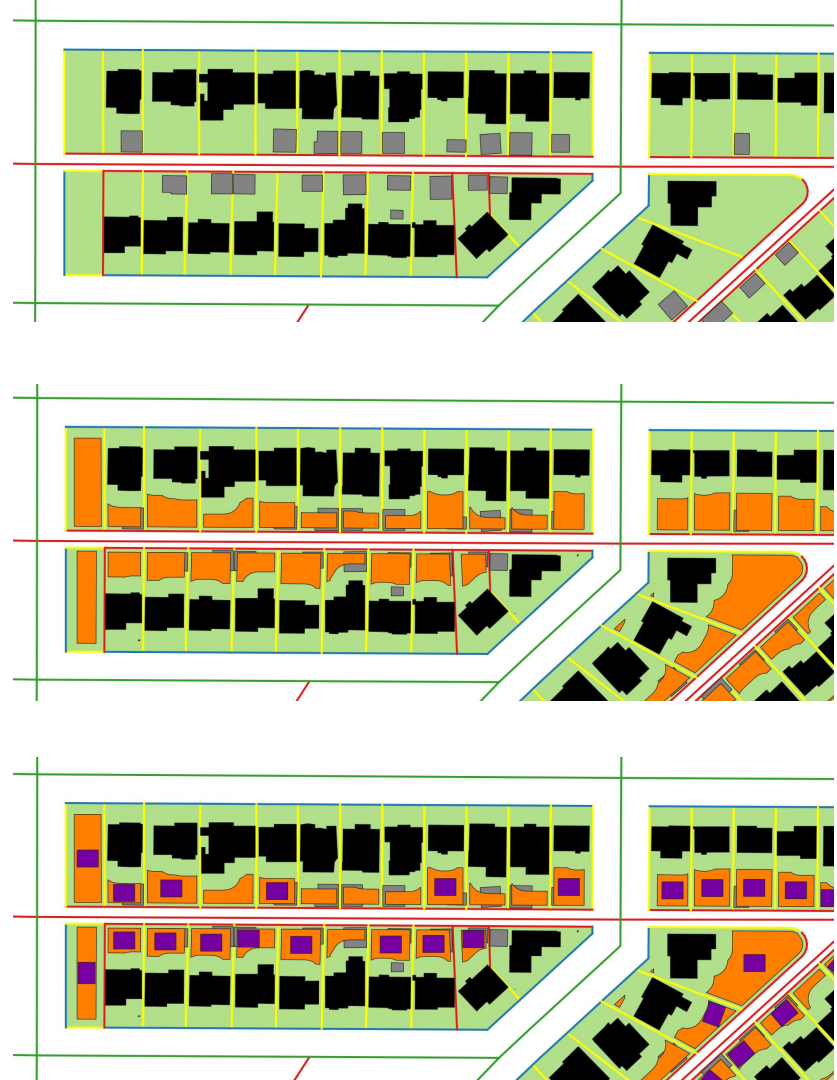


<https://yusenrong46-afk-burnaby-verificat-dashboardsreamlit-app-rcvryj.streamlit.app/~/+/>

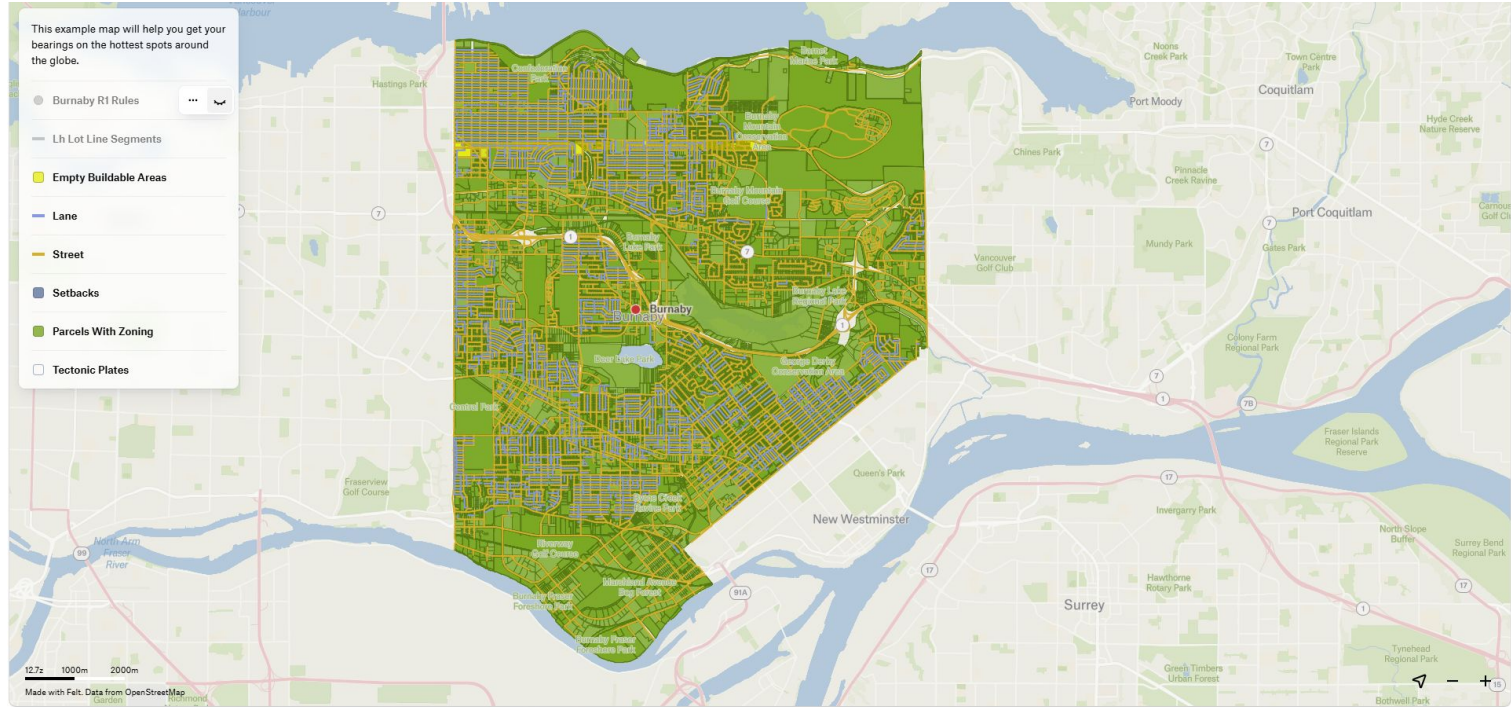
GIS Rule Application: Jamie

1. Label the largest building on each lot as **Primary**, others as **Accessory**
2. Subtract the **Primary** (with min distance buffer) from buildable area
3. Fit the footprint:
 - Translational steps 0.5 m
 - Rotational steps of 10 deg

Implemented for Burnaby and Calgary using common Python scripts.



Dashboard: Sasivimol



Dashboard Overview

<https://felt.com/map/Burnaby-LU239AptVQxCLFQelaFixxD?loc=49.275074,-122.962785,17.05z>

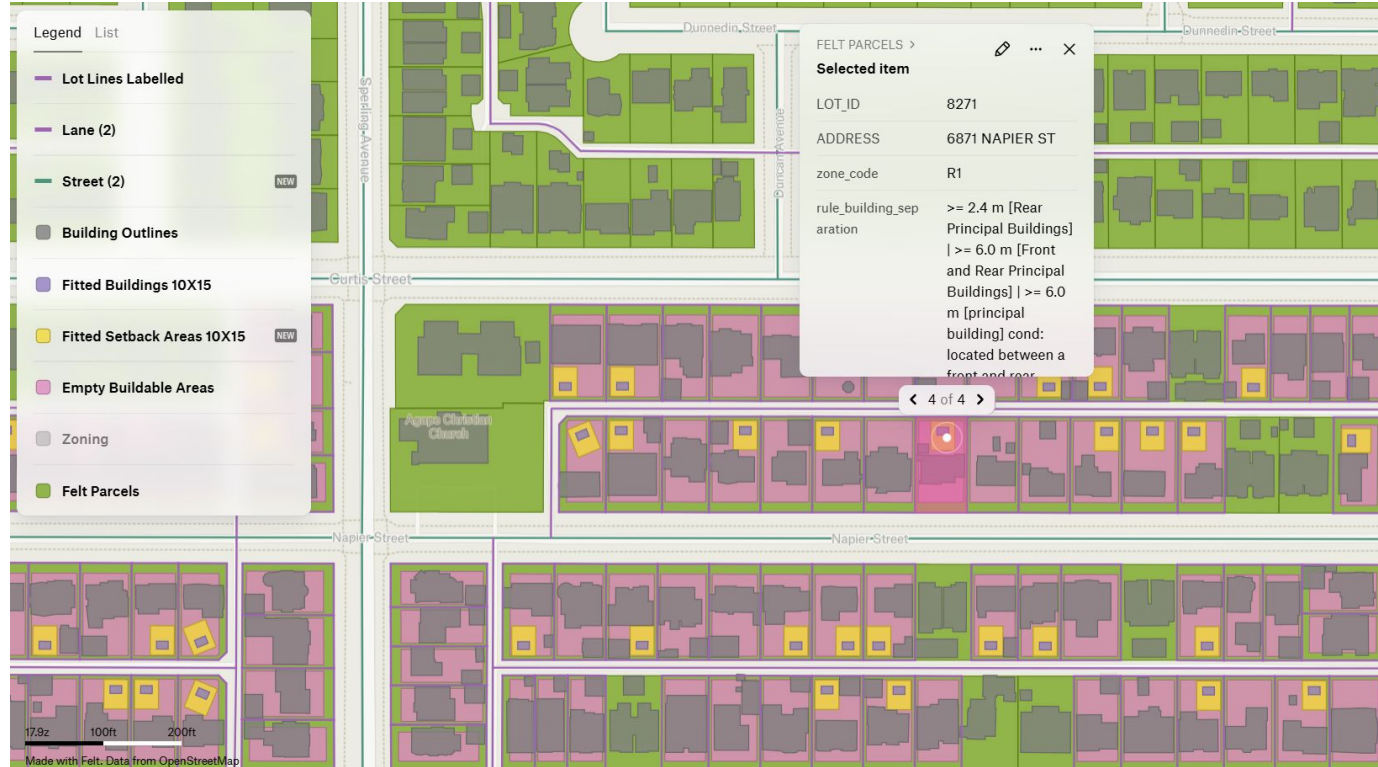
Dashboard: Sasivimol

Current progress:

- All layers display correctly
- Hover tooltip on parcels shows

Challenges:

- Rule attributes in tooltip need cleaner formatting



Next Steps

1. Continue validating the generalized extraction process across Calgary, Vancouver, and Burnaby.
2. Improve evidence quality so fewer rules stay stuck in review.
3. Run scripts on Calgary data to test robustness.
4. Fine-tune GIS scripts to handle edge-cases.
5. Build a Calgary dashboard.

Work Summary

Team Member	Primary Tasks	Hours	Team Member	Primary Tasks	Hours
Jamie	1. Built QGIS scripts for buildable areas accounting for existing structures 2. Built QGIS scripts for building footprint generation and footprint fitting 3. Refined scripts based on GMT feedback	33	Thomas	1. Fixed over-rejection issue — correct but out-of-scope rules moved to "not used" instead of rejected 2. Added candidate-evidence comparison and evidence repair support 3. Updated Streamlined verification dashboard	40
Bee	1. Built spatial join pipeline to combine parcel, zoning, and rule files into a single Felt-ready GeoJSON 2. Uploaded and configured all layers in Felt with working hover tooltip showing zoning rules per parcel	32	Zihao	1. Upgraded to Pipeline 10 — generalized graph/RAG extraction, tested on Burnaby (60 rules) and Calgary (137 rules) 2. Reduced API cost via local block pre-filtering before LLM calls	40